Sprint 5 Report

Sprint 5 Demo: https://youtu.be/Cs6vqP325M0

What's New (User Facing)

- New UI for the Community Analysis Dashboard
- Redesigned UI for the Community page
- Redesigned UI for the Post Editor
- New testing/deployment plan
- Implemented backend features

Work Summary (Developer Facing)

This semester, our team turned the project's goals into four main objectives. First, we redesigned the community page and the editor interface to give teachers a calm, focused experience, organizing features so they are easy to find without overwhelming the user. Second, at the client's request, we designed a new community analysis dashboard that lets administrators monitor community activity and better understand how teachers are using the platform. Third, we developed a testing and deployment plan that gives the client a simple, reliable process for merging our changes into the production environment without disrupting existing functionality. Finally, we continued implementing the backend. Because the community backend was not carried over from the platform's previous version, we have been rebuilding it so that every front-end feature has working functionality behind it.

Unfinished Work

Several pieces of the project are still in progress. The largest is the backend migration: we are continuing to move community functionality over from the platform's previous version, since much of it was not carried forward and has to be rebuilt to work with the current system. At the same time, we are connecting our redesigned community page and editor interface to that backend. Many of the new UI components are complete on the front end but don't yet have live data or full functionality behind them. Once the backend migration is finished, we will complete this integration, test the full workflow from end to end, and then follow our testing and deployment plan to merge the changes into production.

Completed Issues/User Stories

1. Community Page Implementation

User Story: As a user, I want a more organized and intuitive Community page so that I can easily navigate posts and interact with other users.

Acceptance Criteria:

- Create UI mockups for the Community page.
- Include layouts for browsing posts and accessing community features.
- Explore multiple design options to improve navigation and usability.

2. Admin Analytics Dashboard Mockups

User Story: As an administrator, I want a dashboard that visually displays platform statistics so that I can easily monitor user activity and engagement.

Acceptance Criteria:

- Create UI mockups for the Admin Analytics Dashboard.
- Include visualizations for platform statistics, such as posts and user activity.
- Explore different layouts for displaying analytics clearly.

Incomplete Issues/User Stories

1. Community Page Implementation

User Story: As a user, I want an updated Community page so that I can browse and interact with posts through an improved interface.

Acceptance Criteria:

- Implement the approved Community page design.
- Ensure users can browse and interact with posts.
- Verify that the updated interface works with existing backend functionality.

2. Admin Analytics Dashboard Integration

User Story: As an administrator, I want to view accurate platform statistics so that I can monitor activity and make informed decisions.

Acceptance Criteria:

- Connect dashboard visualizations to the development database.

- Display accurate statistics for posts, users, and engagement.
- Test that dashboard data loads and updates correctly.

Code Files for Review (Located in the adapt-ai azure repository)

We cannot show code here as the client's repository is private.

Retrospective Summary

Here's what went well:

- **Client Alignment & Technical Coordination:** Held multiple productive calls with Freeman (CAIRE tech lead) to establish and align on a thorough, standardized testing and deployment strategy for our community features.
- **Continuous Design Refinement:** Maintained weekly feedback loops with the professors and the CAIRE team, using their guidance to iteratively create and refine multiple UI mockups and enhance the overall user experience.
- **Backend Migration & Feature Preparation:** Built upon our previous sprint's work by successfully migrating existing backend functionality and preparing new community capabilities for seamless integration.
- **Reproducible Local Testing:** Standardized our team's local Docker database environment, allowing all team members to develop, test schema additions, and verify role-based permissions locally without external bottlenecks.

Here's what we'd like to improve:

- **Navigating Access Constraints:** Address ongoing delays caused by waiting for direct access to the shared development database, which restricted our ability to run live cloud integration tests earlier in the sprint.
- **Approval Process Bottlenecks:** Streamline communication around the lengthy review and approval cycle with the client team to prevent pull requests and deployment milestones from stalling.
- **Development Continuity:** Minimize downtime between receiving design approvals and executing code merges so feature momentum remains steady across team members.

Here are changes we plan to implement in the next sprint:

- **Finalize CI/CD Testing Pipeline:** Complete the implementation of the automated migration and ephemeral cloud testing pipeline in collaboration with the CAIRE technical team.

- **Apply Final Feature & Database Changes:** Execute the finalized community feature and database updates using our verified local development environment to ensure clean migrations.
- **Submit Deployment-Ready Pull Request:** Open a comprehensive, deployment-ready PR with full automated test coverage and an ephemeral preview link for review and approval by the professors and the CAIRE team.
- **Prepare for Live Deployment:** Coordinate the official merge into the main development branch once all stakeholder approvals are secured, transitioning the community module to active users before the project concludes.